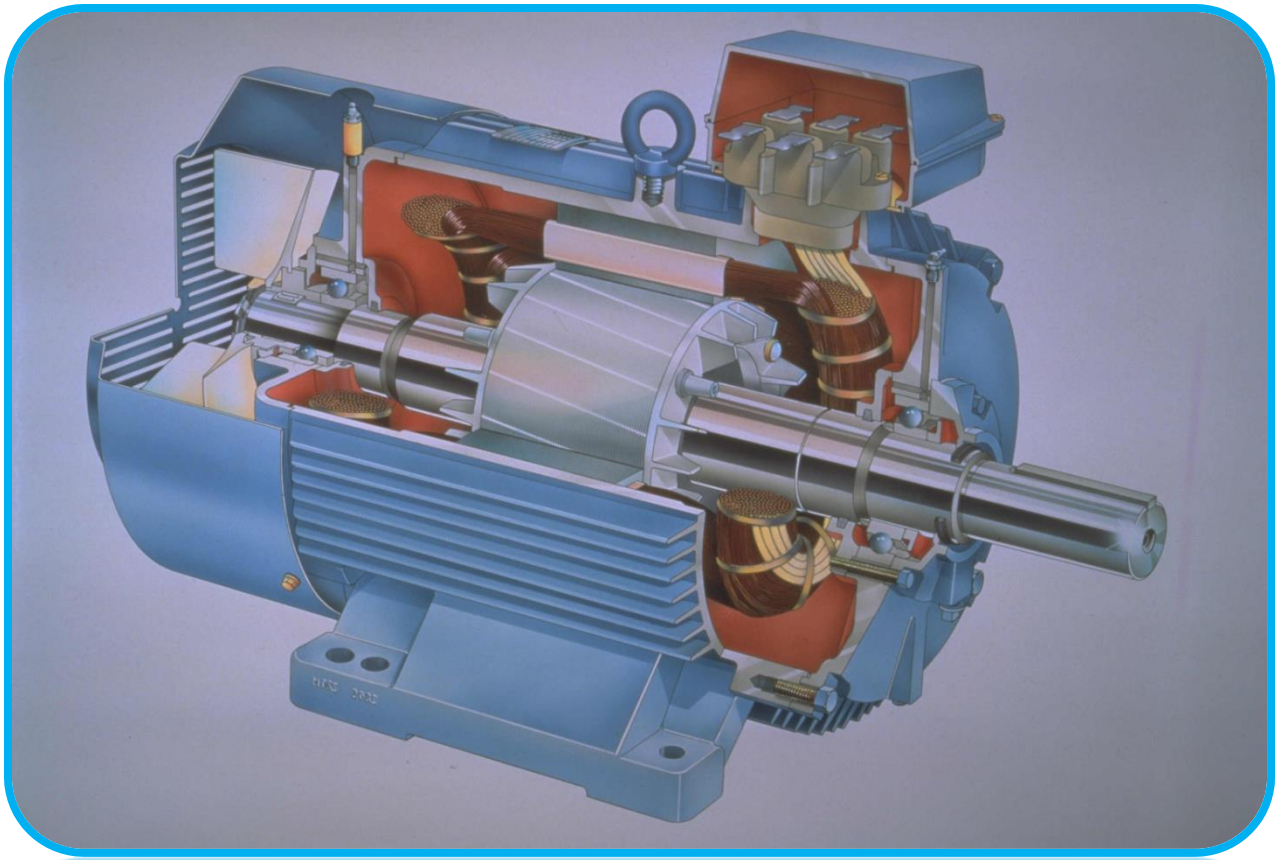

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Module IV
Induction motor
ALLPPT.com _ Free PowerPoint Templates, Diagrams and Charts

Three phase induction Motor



Starting of Induction Motor

- A **starter** is **needed** for a three phase **induction motor** because at t he time of **starting**, if **induction motor** is started directly, it will dr aw a huge amount of current which can cause damage to adjoining equipments
- Such a current can be **6 to 8 times of full load or rated current** a nd it can damage the motor winding.
- Why starting current very high in an induction motor???
  - ✓In a 3 phase IM , the magnitude of an induced emf in the rot or circuit depends on the slip of the induction motor.
  - ✓At starting time, the speed of the motor is zero and slip is at its maximum i.e. unity. So magnitude of rotor induced emf is very large at start. As rotor conductors are short-circuited, th e large induced emf circulates very high current through roto r at start.
  - ✓The condition is exact similar to a transformer with short-cir



Starting of Induction Motor

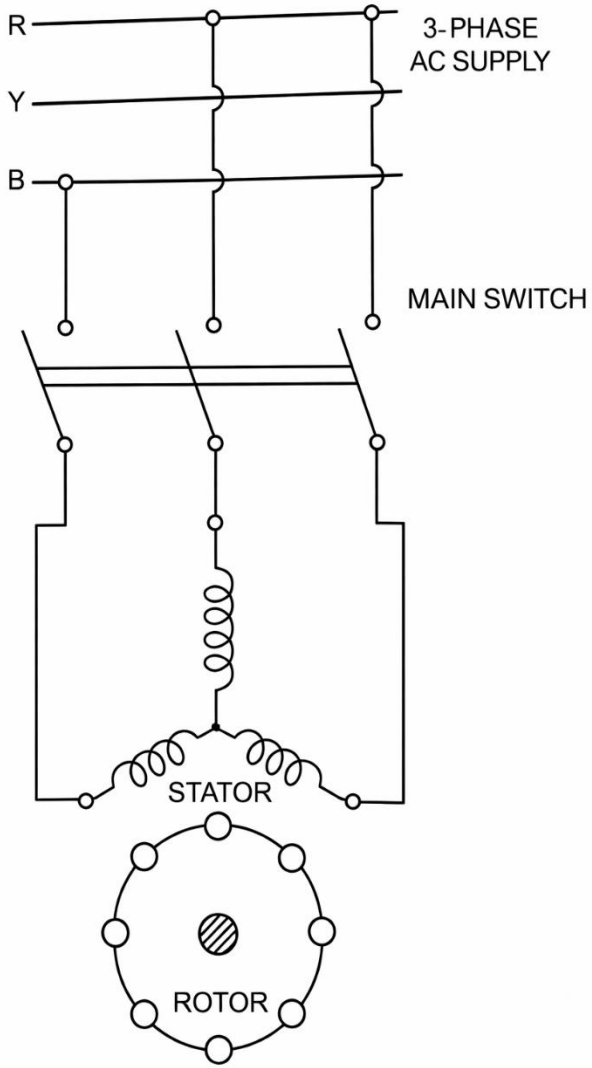
- Reduction in applied voltage to the stator will decrease rotor current. But the starting torque would be greatly decreased.
- $Tst \propto E2^2$
- Starters are required to limit the starting current.

Types

1.Direct On-Line (DOL) Starter
2.Primary Resistor Starting
3.Autotransformer Starter
4.Star-Delta Starter
5.Rotor Resistance Starter

Direct On-Line(DOL) Starter

- Full voltage starting
- Generally used for starting Squirrel cage motors of capacity upto 1.5kW
- Direct switching of motors to the supply mains.



Torque Developed On Starting of Induction Motor By Direct Switching.

Input power to rotor $= T\omega$

Rotor copper loss $= s \times$ input power to rotor

or $3 I^2 R_2 = s T \omega$

where I_2 is the rotor current per phase, R_2 is rotor resistance per phase, T is the torque developed, s is the slip and ω is the supply angular velocity.

$$\text{or Torque developed, } T = \frac{3 I^2 R_2}{s \omega}$$

$$\text{i.e. } T \propto \frac{I_2^2}{s} \quad \text{if rotor resistance is constant}$$

Since rotor current is proportional to stator current I_1, so

$$T \propto \frac{I_1^2}{s}$$

$$\text{or } T = K \frac{I_1^2}{s}$$ where K is any constant

At starting slip, $s = 1$

So starting torque, $T_{st} = K (I_{st})^2$ where I_{st} is the starting current

$$\text{Full-load torque, } T_f = \frac{K I_f^2}{s_f}$$

where I_f is the full-load motor current and s_f is the full-load slip

$$\text{So } \frac{T_{st}}{T_f} = \frac{K I_{st}^{\ 2}}{K I_f^2 / s_f} = \left(\frac{I_{st}}{I_f}\right)^2 s_f$$

$$\text{or } T_{st} = T_f \times \left(\frac{I_{st}}{I_f}\right)^2 s_f$$

Primary Resistor Starter

- Reduced voltage to stator is achieved by connecting resistor in series with each stator lead during the starting period.
- As motor picks up speed, resistors are cut-out in steps & finally short circuited when motor attains normal operating speed.
- If normal supply voltage is V & by using line resistance starter, the voltage is reduced to KV.

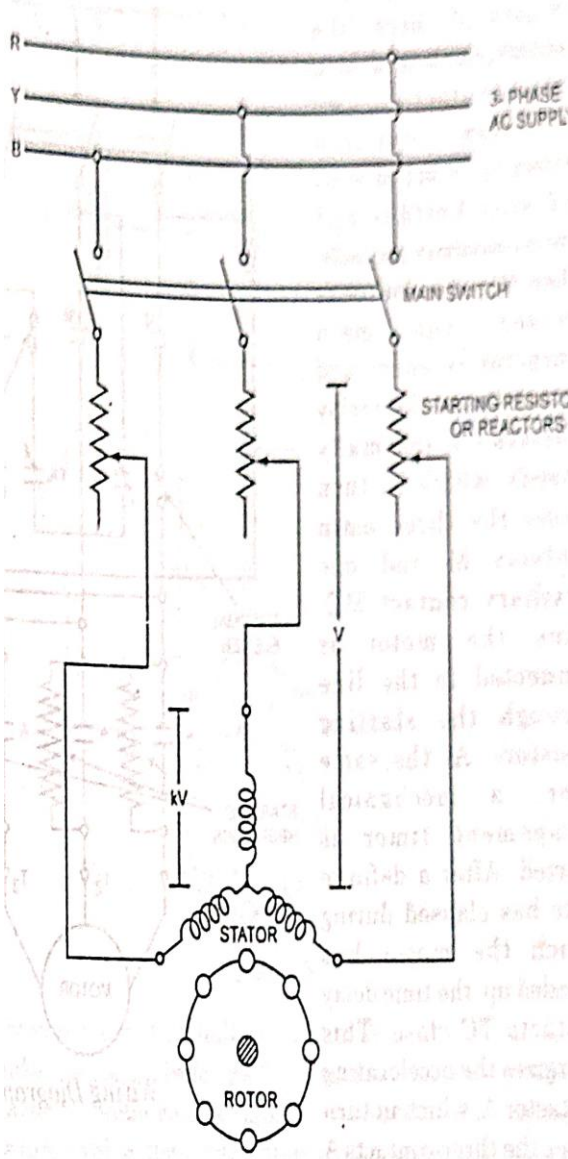
By using primary resistors (Fig. 35.18), the applied voltage/phase can be reduced by a fraction 'x' (and it additionally improves the power factor of the line slightly).

$$I_{st} = x I_{sc} \quad \text{and} \quad T_{st} = x^2 T_{sc}$$

As seen from Art 35.10, above,

$$\frac{T_{st}}{T_f} = \left(\frac{I_{st}}{I_f}\right)^2 . s_f = \left(\frac{x I_{sc}}{I_f}\right)^2 s_f$$

$$= x^2 \left(\frac{I_{sc}}{I_f}\right)^2 s_f = x^2 . a^2 . s_f$$



Advantages. 1. Smooth acceleration.

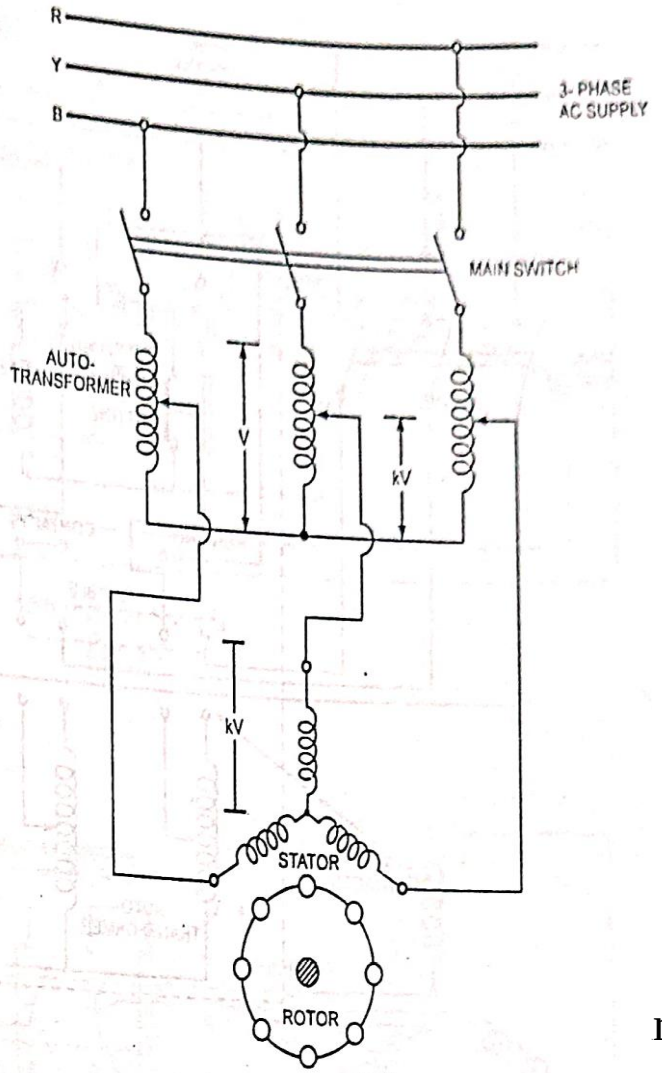
2. High power factor during start.
3. Less expensive than auto-transformer starter in lower output ratings.
4. Closed transition starting.
5. Available with as many as 7 accelerating points.

Disadvantages 1. Resistors give off heat.

2. Low torque efficiency.
3. Starting duration usually exceeds 5 seconds, so needs expensive resistors.

Autotransformer Starter

- Reduced voltage is obtained by taking tappings at suitabl e points from a 3-phase auto -transformer.
- Over-load protection, Under -voltage protection, Thermal over load relays



rter

Autotransformer Starter

protected by thermal overload relays.

Let the motor be started by an auto-transformer having transformation ratio, k. If I_{sc} is the starting current when normal voltage is applied, and applied voltage to stator winding at starting = kV

then motor input current, $I_{st} = k I_{sc}$ ....(8.3)

Supply current = Primary current of auto-transformer

= k × secondary current of auto-transformer $= k^2 I_{sc}$

Starting torque, $T_{st} = T_f \left(\frac{I_{st}}{I_f}\right)^2 \times s_f = T_f k^2 \left(\frac{I_{sc}}{I_f}\right)^2 \times s_f$ ...(8.4)

$= k^2 \times$ Torque obtained by direct switching.





Autotransformer Starter

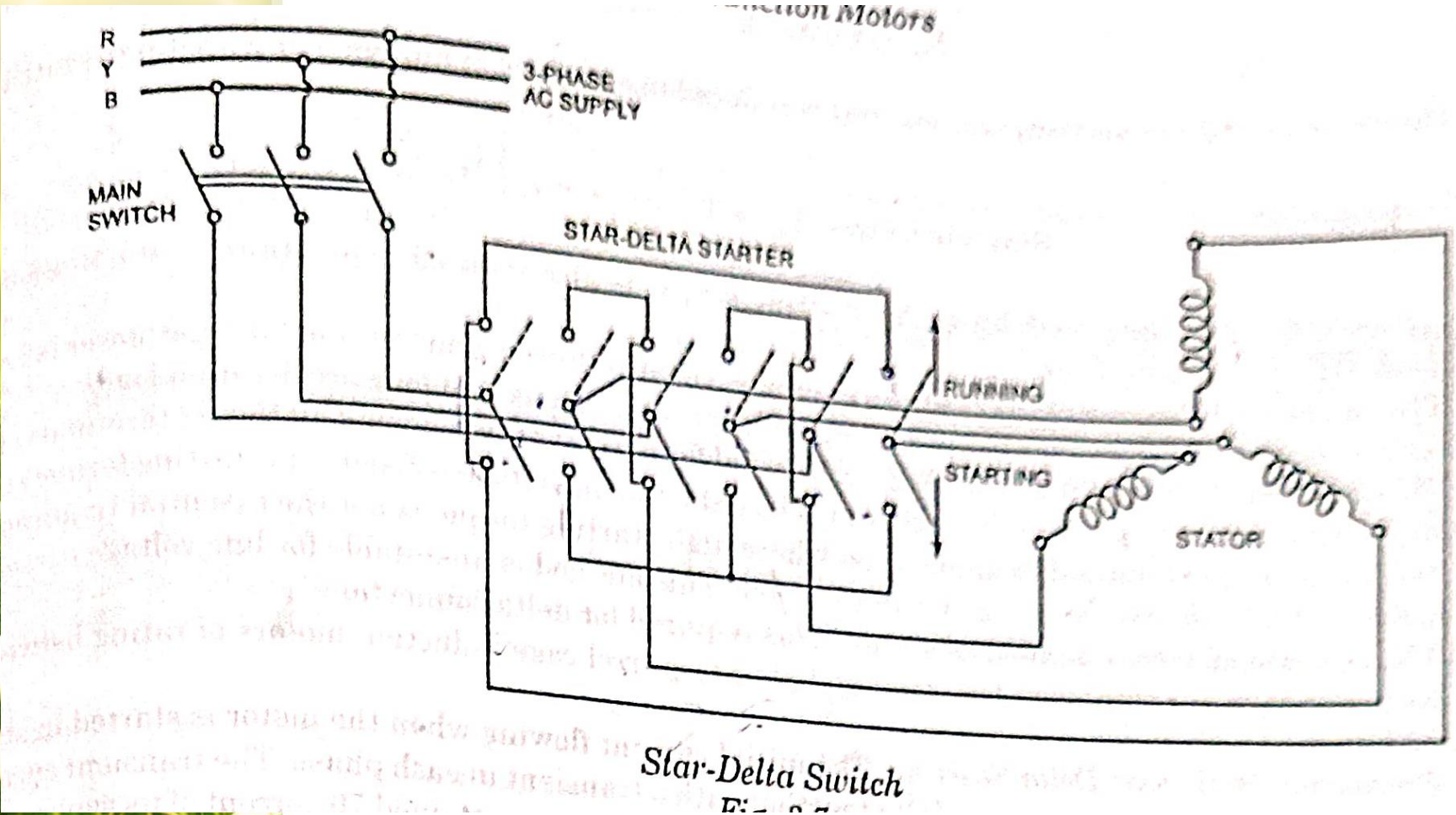
Advantages

- Voltage is reduced by transformation & not by dropping voltage in resistors(as in case of primary resistance starting) → current & power drawn from the supply are reduced.
- Internal losses of the starter itself is low. Highest starting torque per ampere of supply current.
- Adjustment of starting voltage by selection of proper tap on autotransformer.
- Suitability of long starting torque.
- Motor current larger than supply current.
- Can be employed for star connected as well as delta connected motors.

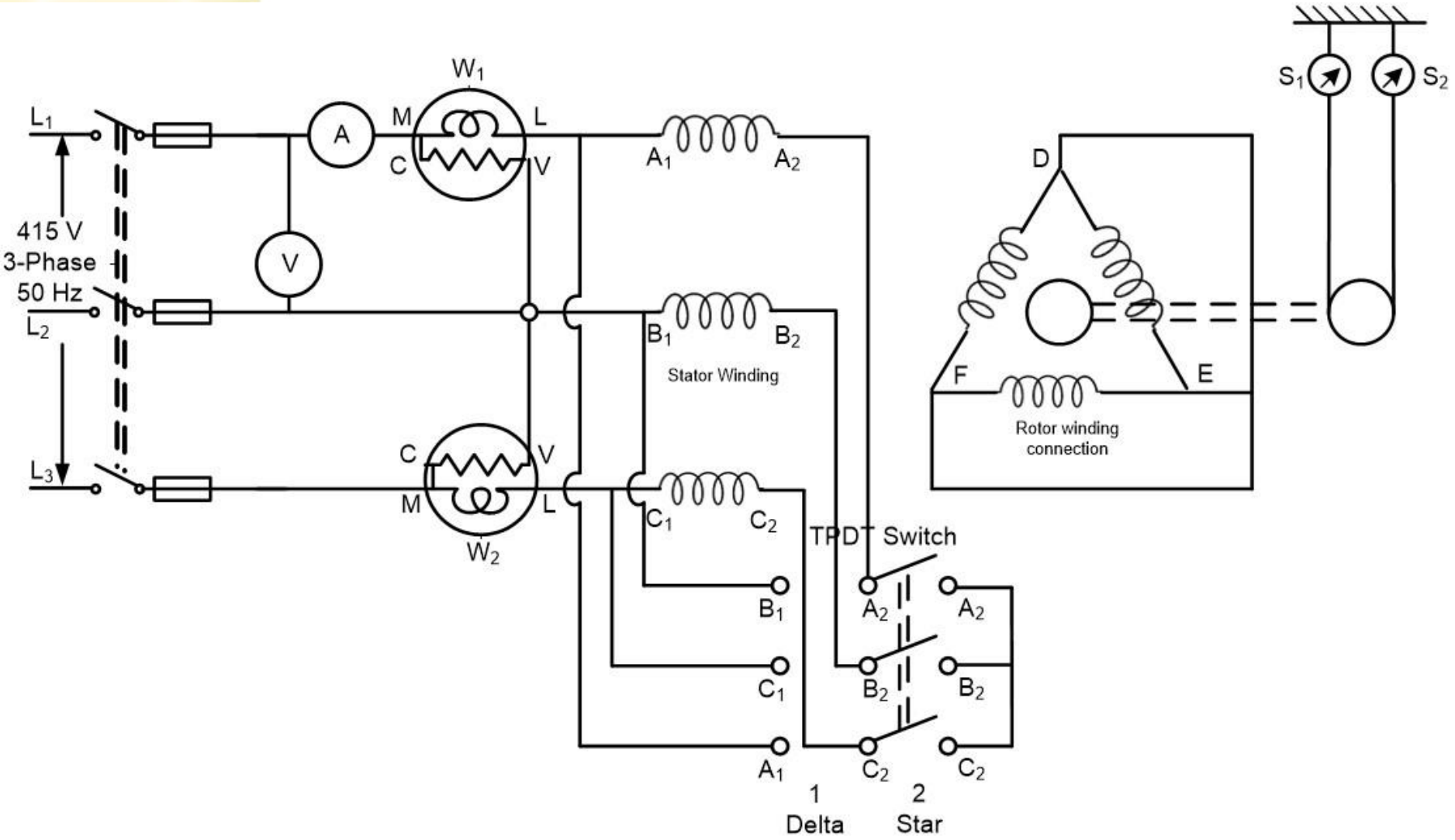
Drawbacks

- Low power factor
- Higher cost in case of lower output rating motors.

Star-Delta Starter



Star-Delta Switch


W1
L1
A
M
L
C
V
A1
A2
S1
S2
D
415 V
3-Phase
50 Hz
V
L2
B1
B2
Stator Winding
F
E
Rotor winding
connection
C
V
L3
M
L
C1
C2
W2
TPDT Switch
B1
A2
A2
C1
B2
B2
A1
C2
C2
1
Delta
2
Star

- Principle - **Voltage across each phase winding of a 3-phase star connected winding is 57.7%(ie, $1/\sqrt{3}$) of the line voltage** whereas same winding connected in delta will have full line voltage across each phase windings.
- Star-Delta starter $\rightarrow$ connects stator winding in star across the supply at starting instant.

Since at starting instant, the stator windings are connected in star, so voltage across each phase winding is reduced to $\frac{1}{\sqrt{3}}$ of line voltage and, therefore, starting current per phase becomes equal to $I_{sc}/\sqrt{3}$.

Starting line current by connecting the stator windings in star at the starting instant = starting motor current per phase = $I_{sc}/\sqrt{3}$.

Starting line current by direct switching with stator windings connected in delta = $\sqrt{3}\ I$

$$\therefore \frac{\text{Line current with star-delta starting}}{\text{Line current with direct switching}} = \frac{I_{sc}/\sqrt{3}}{\sqrt{3}\ I} = \frac{1}{3}$$

Star-Delta Starter

- By star-delta starting, line current is reduced to one-third of line current with DOL switching.
- Starting torque,

Starting torque, $T_{st} = T_f \left(\frac{I_{sl}}{I_f} \right) s_f = T_f \left[\frac{I_{sc}/\sqrt{3}}{I_f} \right]^2 s_f = \frac{1}{3} T_f \left(\frac{I_{sc}}{I_f} \right)^2 s_f$

itching, the starting torque is also reduced to one-thi



Star-Delta Starter

Advantages

- Simple
- Cheap
- Efficient (∵ no power is lost as in case of primary resistance start ing)
- Suitable for high inertia & long acceleration loads.

Drawbacks

- Motor needs to be delta connected for normal operation. Hence 6 terminals of the stator winding are to be brought out.
- Reduction in voltage is fixed.
- Starting torque is low.
- Limited to application where high starting torque is not required.
- Unsuitable for line voltage exceeding 3000V as excess stator tur ns required for delta connection.



Rotor Resistance Starter

- Starting of slip-ring induction motor
- Full line voltage is applied across stator
- Variable resistance is added in each phase of rotor ci rcuit.
- Adding external resistance to the rotor circuit → Red uces starting current Increases starting torque
- Improves power factor
- At starting moment → Complete external resistance i s included in the rotor circuit.
- As motor accelerates → external resistance is cut-out in steps.
- When motor runs at normal speed → rotor winding is short-circuited through slip-rings.

Rotor Resistance Starter

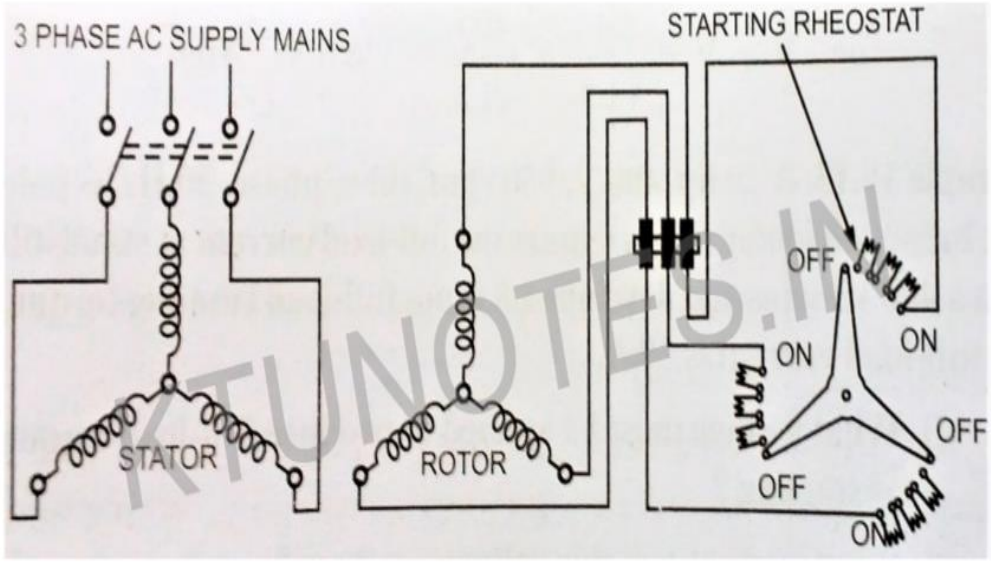


Figure 20 : Rotor Resistance Starter

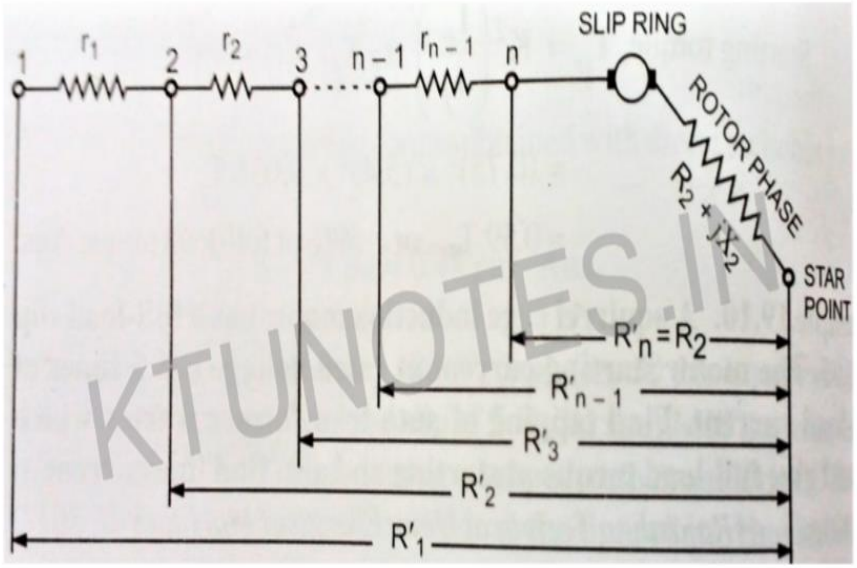


Figure 21 : Rotor resistance starter(Circuit)

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